

CLASS XI COMPUTER SCIENCE (083)

COMPUTER SYSTEM OVERVIEW

Session 2026–27 • Half-Yearly Preparation • Notes + Concepts • No MCQs

Source note: No school PDF/notes for this chapter were attached. These notes therefore use the official CBSE Class XI Computer Science (083) 2026–27 syllabus for the Computer Systems and Organisation topics corresponding to Computer System Overview. School-specific wording may differ.

1. Computer System

Computer: An electronic programmable machine that accepts data/instructions as input, processes them according to instructions, stores data/results and produces output.

Computer system: A complete combination of hardware and software working together to perform useful tasks.

Basic cycle: Input → Processing → Output, with storage used to retain data, instructions and results.

2. Hardware and Software

Hardware: Physical/tangible parts of a computer system, such as CPU, keyboard, monitor, RAM and storage devices.

Software: Programs/instructions that tell hardware what operations to perform.

Hardware	Software
Physical components	Programs/instructions
Can be physically touched	Cannot be physically touched
CPU, keyboard, RAM, SSD	Operating system, browser, word processor

3. Input Devices

An **input device** is hardware used to enter data or instructions into a computer.

- **Keyboard:** enters text, numbers and commands.
- **Mouse:** pointing/selecting device used to interact with graphical interfaces.
- **Scanner:** converts physical documents/images into digital form.
- **Microphone:** provides audio input.
- **Webcam:** provides image/video input.

4. Output Devices

An **output device** presents processed information/results to the user.

- **Monitor:** visual output.
- **Printer:** physical printed output.
- **Speakers/Headphones:** audio output.
- **Projector:** enlarged visual display.

5. CPU — Central Processing Unit

The **CPU** is the main processing component of a computer. It executes instructions and performs the operations required to process data.

ALU (Arithmetic Logic Unit): performs arithmetic and logical/comparison operations.

Control Unit (CU): controls and coordinates instruction execution and data movement.

Registers: very small, very fast storage locations used during CPU processing.

Remember: ALU = calculate/compare • CU = control/coordinate • Registers = temporary working storage.

6. Memory

Memory stores data, instructions and results. For this chapter, distinguish **primary memory**, **cache memory** and **secondary memory**.

Memory type	Concept	Examples / characteristics
Primary	Used directly for active processing.	RAM and ROM; closely associated with CPU operation.
Cache	Very fast memory used to reduce access time for frequently accessed data/instructions.	Small and fast; located close to CPU.
Secondary	Long-term storage.	SSD, HDD, pen drive, memory card, optical storage.

7. RAM and ROM

RAM (Random Access Memory): working memory used by currently running programs/data; generally volatile.

ROM (Read Only Memory): non-volatile memory that retains stored information when power is removed; commonly associated with firmware/start-up instructions.

Volatile: contents are lost when power is removed. **Non-volatile:** contents are retained without continuous power.

8. Units of Memory

A **bit** is a binary digit that can represent 0 or 1. A **byte** consists of 8 bits.

Unit	Meaning
bit	Binary digit (0 or 1)
Byte (B)	8 bits
KB	Kilobyte
MB	Megabyte
GB	Gigabyte
TB	Terabyte
PB	Petabyte

Exam caution: If a numerical conversion is asked, follow the KB/MB/GB convention specified by your teacher or question paper.

9. Types of Software

The official 2026–27 CBSE syllabus groups software into system software, programming tools/language translators and application software.

A. System Software

System software manages computer resources and provides a platform/services for applications.

Operating System: manages resources and provides user/application services.

System Utilities: tools for maintenance, management, protection or optimization tasks.

Device Drivers: software that enables the operating system to communicate with specific hardware devices.

B. Programming Tools and Language Translators

Assembler: translates assembly-language instructions into machine-level code.

Compiler: translates a program into a lower-level form before execution.

Interpreter: processes/translates a program through interpretation as it runs.

C. Application Software

Application software is designed for user-oriented tasks, such as word processing, web browsing, presentations, media playback or education.

10. Operating System (OS)

An **Operating System** is system software that manages computer hardware/resources and provides services and an interface for users and applications.

Major functions of an OS:

- **Process/program management:** manages running programs and processing resources.
- **Memory management:** manages allocation and use of memory.
- **File management:** organizes files/folders and controls access to stored data.
- **Device management:** coordinates hardware devices, commonly using drivers.
- **User interface:** provides a way for users to interact with the computer.
- **Security/access control:** helps control access to system resources and data.

11. OS User Interface

GUI (Graphical User Interface): interaction through windows, icons, menus, buttons, pointers and other graphical elements.

CLI (Command-Line Interface): interaction by entering text commands.

GUI	CLI
Graphical elements	Text commands
Often easier for beginners	Requires knowledge of commands
Mouse/touch + keyboard commonly used	Keyboard is central

12. How Everything Connects

User → Input → CPU/Memory → Output

The operating system manages resources and provides services/interface. Secondary storage keeps programs and data for long-term use.

13. Must-Know Definitions

Term	Definition
Computer System	Hardware and software working together to input, process, store and output information.
Hardware	Physical components of a computer.
Software	Programs/instructions that control computer operations.
CPU	Central Processing Unit; main processing component.
ALU	Performs arithmetic and logical operations.
Control Unit	Controls and coordinates instruction execution.
Input Device	Hardware used to enter data/instructions.
Output Device	Hardware used to present results.
RAM	Volatile working memory used by active programs/data.
ROM	Non-volatile memory that retains stored information.
Operating System	System software that manages resources and provides services/interface.
Device Driver	Software enabling communication between OS and a hardware device.
Compiler	Translator that compiles a program before execution.
Interpreter	Processes/translate a program through interpretation while it runs.

14. What You Should Understand Before Questions

- Hardware vs software.
- Input vs output devices.
- CPU, ALU, Control Unit and registers.
- Primary vs cache vs secondary memory.
- RAM vs ROM and volatile vs non-volatile memory.
- Bit, byte and the sequence KB → MB → GB → TB → PB.
- System software vs application software.
- Operating system, utilities and device drivers.
- Assembler vs compiler vs interpreter.
- Main functions of an operating system.
- GUI vs CLI.

15. Source

Official CBSE Class XI Computer Science (083), 2026–27 syllabus — Unit 1: Computer Systems and Organisation. The syllabus explicitly covers basic computer organisation, hardware/software, input/output devices, CPU, primary/cache/secondary memory, memory units, software categories, language translators, OS functions and OS user interface.

MCQs intentionally omitted. We will first understand the chapter and make the notes solid; questions will come after the concepts are understood.